A CASE-STUDY BASED REPORT

ON

Fake News Detection Using NLP, TF-IDF and BERT

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by

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INDEX

Abstract

Introduction.....

Background.....
Motivation for the Project.....
Overview of the Work Carried Out.....
Scope and Responsible-Use Statement.....

Problem Statement.....

Case-Study Scenario.....
Central Case-Study Question.....
Formal Problem Definition.....
Objectives that Guide the Study.....
Assumptions and Limitations.....

Dataset and Exploratory Data Analysis

About the Dataset.....
Purpose of EDA.....
Preparation Steps Actually Performed.....
EDA Exploration - Implementation.....
Summary of EDA Findings.....

Algorithm Selection and Implementation.....

Conclusion

Signature of the Team members

A CASE-STUDY BASED REPORT

A Case Study on Fake News Detection Using NLP, TF-IDF, and BERT Models

Abstract

Fake news is false or misleading information presented as news and can spread rapidly through online platforms. This case study describes a machine-learning and natural-language-processing system that classifies news text as Fake or Real. The implementation uses a dataset of 2,000 records, text cleaning, TF-IDF feature extraction, Logistic Regression, Linear Support Vector Machine (SVM), and a fine-tuned BERT model. Exploratory data analysis includes class distribution, missing-value and duplicate checks, text length and word-count analysis, feature correlation, and a simple sensationalism score. The Logistic Regression and Linear SVM evaluation outputs in the provided notebook report 100% accuracy, precision, recall, and F1-score on a 400-record test split. The Gradio interface accepts a news title and article and returns a Fake/Real prediction with a confidence estimate using the Logistic Regression pipeline. These predictions are intended to support news analysis, not replace independent fact-checking.

1. Introduction

1.1 Background

Digital news and social platforms allow information to reach large audiences quickly. The same speed can help false or misleading claims spread before they are checked. Natural Language Processing (NLP) can prepare and analyze textual content, while supervised machine-learning models can learn patterns associated with labeled examples of fake and real news.

1.2 Motivation for the Project

The project aims to make an initial screening of news content faster and easier. It also compares traditional TF-IDF-based classifiers with a contextual Transformer approach, BERT, to illustrate differences in model design and computational requirements.

1.3 Overview of the Work Carried Out

- Loaded the Fake_News_Detection_Dataset_2000.csv dataset in Google Colab.
- Checked dataset shape, data types, missing values, duplicate records, and class distribution.
- Combined the news title and article text, cleaned the text, and derived text-related features.
- Split the labeled data into training and testing sets using an 80:20 split with stratification.
- Applied TF-IDF with unigrams and bigrams, then trained Logistic Regression and Linear SVM.
- Prepared and fine-tuned a BERT sequence-classification model for two epochs.
- Built a Gradio interface for interactive predictions using the Logistic Regression model and TF-IDF vectorizer.

1.4 Scope and Responsible-Use Statement

The current implementation classifies the text entered by the user; it does not independently verify the story against authoritative news sources. A model prediction or confidence value is not proof that an article is true or false. Users should verify important claims using reliable sources and human fact-checking.

2. Problem Statement

2.1 Case-Study Scenario

A reader encounters a news headline or article online and wants a quick initial indication of whether its textual patterns resemble the Fake or Real examples in a labeled dataset. Manual checking can take time, especially when many articles need to be reviewed.

2.2 Central Case-Study Question

Can an NLP-based classification pipeline use the text of a news title and article to classify a record as Fake or Real, and how do TF-IDF-based classifiers compare with a BERT-based approach in this project?

2.3 Formal Problem Definition

Given a labeled dataset of news records, learn a function that maps the cleaned text of a title and article to one of two classes: Fake News (label 0) or Real News (label 1). Evaluate predictions on held-out test data using accuracy, precision, recall, F1-score, and a confusion matrix.

2.4 Objectives that Guide the Study

- Inspect and prepare the news dataset for classification.
- Use NLP text cleaning to remove URLs, HTML tags, punctuation, and extra whitespace and convert text to lowercase.
- Convert text into numerical features with TF-IDF.
- Train and evaluate Logistic Regression and Linear SVM models.
- Fine-tune a BERT model as a contextual language-model approach.
- Provide an interactive Gradio interface for a title/article input and a prediction output.

2.5 Assumptions and Limitations

The implementation assumes that the dataset labels are correctly assigned and that the training examples are representative of future inputs. The uploaded notebook reports perfect scores for Logistic Regression and SVM; such results should be interpreted cautiously and verified on independent, realistic data. The recorded model comparison table includes Logistic Regression and Linear SVM, but does not include a BERT test-score row. The Gradio prediction function uses Logistic Regression, not BERT. The system also relies on text alone and may not capture source credibility, images, events, or external evidence.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The uploaded notebook loads Fake_News_Detection_Dataset_2000.csv with 2,000 records and 18 original columns. The original fields include News_ID, Title, Text, Source, Subject, Author, PublishedDate, several text-derived features, and class. The project creates a label column from class, with 0 mapped to Fake News and 1 mapped to Real News. The PPT presents a simplified four-field view—serial_number, title, text, and label—as the core schema. For the implemented text classifier, the principal inputs are Title and Text, and the target is label/class.

The notebook outputs show 1,000 records in each class (50% Fake and 50% Real), with zero missing values and zero duplicate records reported in the initial dataset checks.

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) is used to understand dataset quality, check whether the target classes are balanced, inspect text length and word count, examine differences between Fake and Real examples, and explore relationships among selected numerical features before model training.

3.3 Preparation Steps Actually Performed

- Loaded the CSV and inspected its dimensions, columns, and data types.
- Checked missing values and duplicate rows.
- Created a label column from the class column and mapped 0 to Fake News and 1 to Real News.
- Filled missing Title and Text values with empty strings and combined the two text fields.
- Cleaned text by lowercasing, removing URLs and HTML tags, removing punctuation, and normalizing whitespace.
- Created or analyzed text length, word count, punctuation count, URL count, exclamation count, question count, and a simple sensationalism score.
- Used a stratified 80:20 train-test split, producing 1,600 training records and 400 testing records.

3.4 EDA Exploration – Implementation

The notebook generates a class-distribution count plot; text-length and word-count summaries; Fake-versus-Real boxplots for text length and word count; a correlation heatmap for the label and selected text features; feature histograms and boxplots; and a comparison of the derived sensationalism score by news type.

3.5 Summary of EDA Findings

- Dataset size in the notebook: 2,000 rows and 18 original columns.
- Class balance: 1,000 Fake and 1,000 Real records.
- Missing values reported: 0; duplicate records reported: 0.
- Reported mean text length: approximately 261 characters.
- Reported mean word count: approximately 38 words.
- The notebook's simple sensationalism score ranges from 0 to 2, with a mean of 1.

4. Algorithm Selection and Implementation

4.1 NLP Preprocessing

The title and article are combined into one text string. The clean_text function converts the text to lowercase, removes URLs and HTML tags, removes punctuation, collapses repeated whitespace, and trims leading/trailing spaces.

4.2 TF-IDF Feature Extraction

TfidfVectorizer is configured with max_features=10000, English stop-word removal, and ngram_range=(1, 2), which allows single words and two-word sequences. The vectorizer is fitted on training text and then applied to test text. The recorded output shows 417 TF-IDF features for this dataset.

4.3 Logistic Regression

Logistic Regression is trained on the TF-IDF training matrix with max_iter=1000 and random_state=42. It is the classifier used by the Gradio prediction function. The notebook reports 100.00% accuracy, precision, recall, and F1-score on the 400-record test set, with a confusion matrix of [[200, 0], [0, 200]].

4.4 Linear Support Vector Machine (SVM)

A LinearSVC classifier with random_state=42 is trained using the same TF-IDF training features. Its recorded evaluation reports 100.00% accuracy, precision, recall, and F1-score, with a confusion matrix of [[200, 0], [0, 200]].

4.5 BERT

The notebook prepares a text-and-label dataset for BERT, tokenizes the text using bert-base-uncased, and fine-tunes a sequence-classification model for two epochs. The recorded training output reports a training loss of about 0.01425. Although the notebook defines evaluation metrics for BERT, the saved comparison table shown in the provided outputs contains only Logistic Regression and Linear SVM. Therefore, a final BERT test metric is not reported in this case-study text.

4.6 Model Evaluation and Comparison

Accuracy measures the proportion of correct predictions. Precision measures the proportion of predicted positive cases that are correct; recall measures the proportion of actual positive cases identified; and F1-score is the harmonic mean of precision and recall. A confusion matrix summarizes correct and incorrect classifications.

Recorded model comparison (from notebook outputs):

- TF-IDF + Logistic Regression — Accuracy 100.00%; Precision 100.00%; Recall 100.00%; F1-score 100.00%.
- TF-IDF + Linear SVM — Accuracy 100.00%; Precision 100.00%; Recall 100.00%; F1-score 100.00%.
- BERT — final test metrics are not shown in the saved comparison output.

Note: The comparison reproduces the recorded notebook outputs. Because perfect scores can reflect dataset construction or patterns that do not generalize, further validation on an independent dataset is recommended before making claims about real-world performance.

4.7 Gradio Frontend

The Gradio interface is titled “Fake News Detection System.” It provides text boxes for a news title and full article, a CHECK NEWS button, and output fields for the prediction and confidence. The interface calls the predict_news function, which applies the same text cleaning, transforms the input using the TF-IDF vectorizer, predicts with Logistic Regression, and displays the highest predicted class probability as confidence.

5. Conclusion

This case study implements an end-to-end fake news text-classification workflow in Google Colab. The notebook covers dataset inspection, EDA, NLP preprocessing, TF-IDF feature extraction, Logistic Regression and Linear SVM training, BERT fine-tuning, model evaluation, and a Gradio interface. The recorded Logistic Regression and SVM outputs achieve 100% scores on the project test split, but these results should be validated on independent data before drawing conclusions about real-world reliability. The current frontend uses Logistic Regression for its predictions. Future work may include reporting BERT's held-out test metrics, testing on larger and more diverse real-world datasets, adding explainability, and integrating external fact-checking sources.

References / Project Materials

1. ML Abstract: “Fake News Detection Using NLP, TF-IDF and BERT.”

2. Project presentation: “Fake News Detection Using NLP, TF-IDF and BERT.”

3. Google Colab notebook: Frontend_Fake_News_Detection(1).ipynb.

4. Google Colab notebook: Fake_News_Detection_Dataset (EDA notebook).